

File Systems and Storage -- Guided Notes

Unit tp-5.2 | CompTIA Network+ N10-009 Obj. FC0-U71 5.2 | N10-008 FC0-U71 5.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A file system organizes how data is _____, named, and accessed on a storage device. Without a file system, a drive is just a _____ series of bytes. Common Windows file systems include _____ (legacy FAT) and _____ (modern, supports large files and permissions). Linux commonly uses _____.

Question 2

FAT32 has a maximum single file size of _____ GB. NTFS supports files up to _____ TB theoretically. exFAT was designed for _____ drives and supports large files without the 4 GB FAT32 limit. NTFS supports features FAT32 lacks: _____ (file permissions), _____ (access logs), and file-level encryption.

Question 3

A _____ is the directory hierarchy root from which all files are organized. On Windows, drive letters like _____ are the root for each volume. On Linux and macOS the entire filesystem has a single root called _____. Linux uses _____ as the path separator; Windows uses _____.

Question 4

A file _____ (like .txt, .docx, .exe) tells the OS what type of data the file contains and which program to open it with. Extensions are not embedded in the data -- they are just part of the _____ and can be changed. On Windows, the _____ of a file is stored in NTFS metadata, not in the extension.

Question 5

Compression reduces _____ size by encoding repeated patterns more efficiently. _____ compression (ZIP, gzip) reduces size without losing data and can be restored to the original. _____ compression (JPEG, MP3) permanently discards data to achieve greater size reduction. Lossy compression is acceptable for _____ where small quality loss is imperceptible.

Question 6

Encryption converts file data into _____ text that cannot be read without the _____ key. BitLocker (Windows) and FileVault (macOS) are full-disk _____ solutions that encrypt the entire drive. If the drive is lost or stolen, the data cannot be read without the _____ key or recovery code.

Question 7

A _____ is a file or folder shortcut (symbolic link on Linux/macOS, or shortcut .lnk on Windows) that points to another location without copying the data. Deleting the shortcut does not delete the _____ file. On Linux, a _____ link is a direct pointer (inode reference) to the same data, so deleting one does not delete the other as long as another link exists.

Question 8 -- Real World Application

REAL WORLD: A student copies a 5 GB video file from their laptop to a FAT32 USB drive and receives an error saying the file is too large even though the drive has 16 GB of free space. Explain why this error occurs and describe two ways to solve it.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-5-2-file-systems-and-storage/tp-5-2-file-systems-and-storage-answer-key.pdf
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